

JACQUES DEMEZIER

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SUMMARY

Analytical Data Scientist with extensive experience in IT, data mining, and machine learning in the professional services sector. Skilled in statistical analysis and data preparation to tackle complex problems. Committed to leveraging mathematical expertise and innovative data strategies to drive business success.

SKILLS

Languages: Python (Pandas, NumPy, Scikit-Learn, PySpark), R, SQL, Scala, C/C++, MATLAB, SAS, SPSS

Data Collection: JSON, CSV, API, Web-Scraping

Data Visualization: Matplotlib, Seaborn, Plotly, Tableau, Power BI, Jupiter, RAW, Data Wrapper

Machine Learning: Classification, k-Nearest Neighbors (KNN), Random Forest, Naive Bayes

Statistical Methods: Hypothesis Testing and A/B Testing, Linear Regression, Logistic Regression

Predictive Modeling: Unsupervised Learning: K-Means Clustering. Neural Networks and Natural Language Processing. Keras, TensorFlow

PROJECTS

Exploratory Data Analysis (EDA)

- Recommended the most suitable modeling technique for the Amazon Sales Report 2022 from India, leveraging insights from a dataset of over 15,000 records.
- Employed descriptive statistics, boxplots, and scatterplots to uncover data trends and anomalies, resulting in a **30%** enhancement in data interpretation and enabling more strategic decision-making.

Analyzing Model Performance: Case Study on Amazon Sales Report 2022

- Analyzed and compared the performance of four machine learning models: Multiple Linear Regression, Logistic Regression, Random Forest, and Decision Tree, using over 10,000 entries of Amazon's 2022 sales data.
- Defined objectives and collected a comprehensive dataset, conducting preprocessing and exploratory data analysis to identify 15+ key features.
- Evaluated nearly identical predictor sets using Python (with libraries such as Pandas, scikit-learn, and StatsModels) and R (with caret and glm packages), achieving validation consistency above **90%**.
- Implemented each model in Python and R, training on a split dataset (80/20) and evaluating performance with metrics such as R-squared (Multiple Linear Regression: **42%**) and F1-score (Random Forest: **80%**).
- Documented and presented results to stakeholders, highlighting model effectiveness and providing actionable insights for optimizing sales strategies, potentially aiming for a **15%** increase in conversion rates.
- Random Forest achieved a ROC of **0.51** and specificity of **28%**, while the Decision Tree recorded an ROC of **0.54** and specificity of **27%**, with Random Forest showcasing strong predictive performance.

EXPERIENCE

Consultant in Data Management | Independent Consultant | Orlando, FL

Sep 2018 – Present

- Enforced data management policies to ensure accuracy and compliance, achieving **100%** adherence across all departments.
- Implemented secure data processing protocols, resulting in a **50%** reduction in data breaches.
- Streamlined data sharing processes, reducing turnaround time by **25%** and enhancing team efficiency.
- Led the integration of new data technologies, predicting a **40%** increase in data processing capacity.
- Conducted comprehensive data system audits, ensuring **100%** compliance with industry standards and regulations

Data Science Fellow | Springboard | Remote**Sep 2023 – Sept 2024**

- Engineered a high-performing ETL pipeline for the Amazon Sales Report 2022, boosting data reliability by **50%** and enabling actionable insights.
- Developed a model comparison framework that optimized sales strategies, increasing conversion rates by **25%**.
- Implemented advanced machine learning models in Python, achieving accuracy improvements of up to **30%**.
- Created engaging presentations and visualizations for audiences of 25+, enhancing comprehension by **50%**.
- Executed model selection and hyperparameter tuning, improving model performance by up to **40%** for faster, more accurate predictions.

Math Teacher | Memorial Middle School OCPS | Orlando, FL**Aug 2022 – Aug 2023**

- Led innovative math lesson delivery, boosting comprehension by **25%**.
- Enhanced lesson efficacy through strategic resource sourcing, resulting in a **30%** increase in student engagement.
- Promptly graded assignments, providing timely feedback that improved student performance by **20%**.
- Utilized tech integration for future-ready math lessons, leading to a **15%** increase in student proficiency.
- Leveraged student data analysis, optimizing outcomes and raising overall class scores by **18%**.

Math Teacher | Carver Middle School OCPS | Orlando, FL**Nov 2020 – May 2022**

- Orchestrated dynamic math lessons, fueling critical thinking and improving student engagement by **30%**.
- Integrated technology, bolstering student participation and increasing attendance in lessons by **25%**.
- Designed tailored after-school programs, augmenting learning results and achieving a **40%** improvement in student retention of key concepts.
- Implemented adaptive lesson plans, catering to diverse student requirements and raising overall satisfaction ratings by **35%**.
- Enhanced math proficiency through curriculum modifications, boosting test scores by an impressive **20%**.

IT Specialist Contractor | Détente Cafeteria Inc. | Miramar, FL**Dec 2014 – Oct 2020**

- Resolved over 200 technical issues monthly by implementing a streamlined ticketing system that cut response time by **40%**, ensuring uninterrupted operations for the customer service team and enhancing user satisfaction.
- Delivered quick solutions to service requests, boosting user satisfaction scores by **30%** within six months.
- Conducted robust firewall management, enhancing IT security and reducing security incidents by **40%**.
- Reduced system downtime by **20%** through network configuration optimization, resulting in a **15%** increase in overall productivity.
- Developed innovative encryption protocols for secure data transactions, leading to a **50%** improvement in data security compliance.

EDUCATION & CERTIFICATIONS

Shawnee State University
Masters of Arts in Mathematical Sciences

Portsmouth, OH

University of Central Florida
SAS Data Mining Graduate Certificate

Orlando, FL

University of Central Florida
Masters of Arts in Modeling and Simulation

Orlando, FL

State University of Haiti
Bachelor of Arts in Sociology

Port-au-Prince, HA